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A Greedy Approach for a Time-Dependent Scheduling Problem

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Abstract. The aim of the paper is to construct a greedy algorithm for an approximate polynomial-time solution of a single machine time-dependent scheduling problem. In the problem processing time p_j of each job is described by a function of the starting time t of the job, $p_j = 1 + \alpha_j t$, where $\alpha_j > 0$ for $j = 1, 2, \ldots, n$. Jobs are non-preemptable and independent, there are neither ready times nor deadlines, and the criterion of optimality is the total completion time.
There are two main contributions of the paper. Firstly, some theoretical background for proposed algorithm is presented. In this setting useful notions of signatures for given sequences of coefficients of the problem are introduced and their properties are considered. Secondly, these general results are applied to the case of the problem when its coefficients are consecutive natural numbers.

Keywords: single machine; time-dependent scheduling; deteriorating jobs; total completion time

1 Introduction

Scheduling jobs with time-dependent processing times is a new research field in modern scheduling theory. In problems of this kind the processing time of a jobs is depending on the starting time of the job, and this dependency is described by a function. There are known numerous applications of time-dependent scheduling, e.g. in modeling of financial operations, problems of military activities, maintenance or cleaning assignments, etc.

There is available quite large literature of the subject. Most of known results concern a single machine and the makespan [5], [12], [13] or the total completion time [4], [6], [11]. There have been also published results for parallel machines: identical [2], [9] or dedicated [10]. For more details on time-dependent scheduling we refer the reader to [1], [3].

In this paper we consider a single machine scheduling problem with linear time-dependent processing times and the total completion time criterion. Processing times of jobs are in the form of $p_j = 1 + \alpha_j t$ where $\alpha_j \geq 0$, $j =$

$0, 1, 2, \ldots, n$, and t is the starting time of a job. We are interested in minimizing the total completion time $\sum_{j=0}^{n} C_j$, where

$$C_j = \begin{cases} 1, & j = 0 \\ 1 + (1+\alpha_j)C_{j-1}, & j = 1, 2, \ldots, n. \end{cases} \tag{1}$$

The optimization is carried over all permutations of a given sequence $\hat{\alpha}$, where $\hat{\alpha} = (\alpha_0, \alpha_1, \ldots, \alpha_n)$. The main aim of the paper is to present a heuristic for the problem, using *sigmatures* of the sequence $\hat{a} = (a_0, a_1, \ldots, a_n)$, where $a_j = 1 + \alpha_j$ for $j = 0, 1, \ldots, n$.

Due to space limitations we present some our results without proofs; the full version of the paper is presented in the report [7].

2 Preliminary Results

Note that the formula (1) can be written for $j = -, \ldots, n$ as a superposition $C_j = f_j(f_{j-1}(\ldots(f_0(t))\ldots))$ of the functions $f_j(t) = t + p_j(t) = 1 + a_j t$, where $a_j = 1 + \alpha_j$ and task 0 starts at $t = 0$.

Starting at $t = 0$ and expanding the formula $C_j = f_j(f_{j-1}(\ldots(f_1(t))\ldots))$ for $j = 1, 2, \ldots, n$, we obtain $C_0 = 1, C_1 = a_1 C_0 + 1$, $C_2 = a_2 C_1 + 1$, $\ldots$, $C_n = a_n C_{n-1} + 1$. These recurrence relations can be rewritten in the matrix form as follows:

$$\begin{bmatrix} 1 & 0 \ldots & 0 & 0 \\ -a_1 & 1 \ldots & 0 & 0 \\ 0 & -a_2 \ldots & 0 & 0 \\ \vdots & \ldots & & \vdots \\ 0 & 0 \ldots & -a_n & 1 \end{bmatrix} \begin{bmatrix} C_0 \\ C_1 \\ C_2 \\ \vdots \\ C_n \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix},$$

what we will denote in short $A(a)C(a) = d(1)$, where $A(a)$ is the matrix given above, $C(a) = [C_0, C_1, C_2, \ldots, C_n] \in \mathbb{R}^{n+1}$, $d(1) = [1, 1, \ldots, 1]^\top \in \mathbb{R}^{n+1}$.

It is easy to see that if we start at $t = 0$, the coefficient a_0 has no effect on the value of C_j for each j. Therefore [11], given any ordering of the sequence $\hat{a} = (a_0, a_1, \ldots, a_n)$ the best strategy is to set as a_0 the maximal element in this sequence. Hence, from now on, we assume that $a_0 = \max\{a_0, a_1, \ldots, a_n\}$ and we consider only the vector $a = (a_1, a_2, \ldots, a_n)$

Let $\bar{a} = (a_n, a_{n-1}, \ldots, a_1)$ denote the reversed permutation of elements of the sequence a. Note that the components $C_j(a)$ of $C(a)$ and $C_j(\bar{a})$ of $C(\bar{a})$ can be easily calculated. namely,

$$C_j(a) = 1 + \sum_{i=1}^{j} a_i a_{i+1} \ldots a_j \quad \text{and} \quad C_j(\bar{a}) = 1 + \sum_{i=n-j+1}^{n} a_{n-j+1} a_{n-j+2} \ldots a_i,$$

respectively, where $j = 0, 1, \ldots, n$.

We will apply two l_p-norm related measures (see, e.g., [8] for definition of l_p-norm and its properties) to the vector $C(a)$, namely

$$F(a) = \sum_{j=1}^{n} \sum_{i=1}^{j} a_i \cdots a_j \quad \text{and} \quad M(a) = 1 + \sum_{i=1}^{n} a_i \cdots a_n.$$

Note that $\|C(a)\|_1 = \sum_{j=0}^{n} C_j(a) = \sum_{j=0}^{n} \sum_{i=1}^{j} a_i \cdots a_j = F(a) + (n+1)$ and $\|C(a)\|_\infty = \max_{0 \le j \le n} \{C_j\} = M(a)$. (Hence $\|C(a)\|_1 \equiv \sum C_j$ and $\|C(a)\|_\infty \equiv C_{\max}$.)

Let Π_n denote a set of all permutations of the sequence $(1, 2, \ldots, n)$. If $\pi \in \Pi_n$ is any such permutation, then the respective permutation of the given sequence $a = (a_1, \ldots, a_n)$ will be denoted by a_π.

We will refer to minimizing of $F(a)$ as to $F(a)$-*problem*. According to [11], if $a = a_{\pi^*}$ is the optimal solution of the $F(a)$-problem, then $\bar{a} = (a_n, \ldots, a_1)$ is also optimal solution of the $F(\bar{a})$-problem. Moreover, such optimal $a = a_{\pi^*}$ must have V-shape [6], [11]. (This means that the sequence is non-increasing for $i = 1, \ldots, m$, and is non-decreasing for $i = m, \ldots, n$ where $1 < m < n$. We will call such sequences by *V-sequences*.)

Similarly, minimization of $M(a)$ we will call by $M(a)$-*problem*. It is known (see, e.g., [5], [13]) that the $M(a)$-problem has a non-increasing solution.

For a given sequence $a = (a_1, a_2, \cdots, a_n)$, define two *signatures* of a:

$$S^-(a) \equiv M(\bar{a}) - M(a) = \sum_{i=1}^{n} a_1 \cdots a_i \; - \; \sum_{i=1}^{n} a_i \cdots a_n,$$
$$S^+(a) \equiv M(\bar{a}) + M(a) = \sum_{i=0}^{n} a_1 \cdots a_i \; + \; \sum_{i=0}^{n} a_i \cdots a_n.$$

Introduce also the following notation. Given $a = (a_1, \ldots, a_n)$ and any two numbers α, β, let $a^L = (\alpha|a|\beta)$ and $a^R = (\beta|a|\alpha)$ denote the concatenations of α, β and the sequence a in the indicated orders, respectively.

Lemma 1. *For a given sequence a, let $a^L = (\alpha|a|\beta)$ and $a^R = (\beta|a|\alpha)$. Then there hold the equalities*

$$F(\alpha|a|\beta) = F(a) + \alpha\, M(\bar{a}) + \beta\, M(a) + \alpha \cdot a \cdot \beta,$$
$$F(\beta|a|\alpha) = F(a) + \beta\, M(\bar{a}) + \alpha\, M(a) + \alpha \cdot a \cdot \beta,$$

where $\alpha \cdot a \cdot \beta = \alpha \cdot \left(\prod_{j=1}^{n} a_j \right) \cdot \beta$.

Proof. Let $a = (a_1, \ldots, a_n)$, $a_0 = \alpha$, $a_{n+1} = \beta$. Then we have

$$\begin{aligned} F(a_0|a|a_{n+1}) &= \sum_{j=0}^{n+1}\sum_{i=0}^{j} a_i \cdots a_j = F(a) + \sum_{j=0}^{n} a_0 \cdots a_j + \sum_{i=0}^{n+1} a_i \cdots a_{n+1} \\ &= F(a) + a_0\left(1 + \sum_{j=1}^{n} a_1 \cdots a_j\right) + a_{n+1}\left(1 + \sum_{i=1}^{n} a_i \cdots a_n\right) + \\ &+ a_0 a_1 \cdots a_n a_{n+1}, \end{aligned}$$

and the first identity follows. To get the second one, it is sufficient to replace α and β and to note that the last term remains unchanged during this exchange. □

As a corollary from Lemma 1 and definitions of the signatures $S^-(a)$, $S^+(a)$ we obtain general formulae concerning the difference and the sum of values of the functional $F(\cdot)$ on the sequences a^L and a^R.

Lemma 2. *Under the notation of Lemma* 1 *the following identities hold true:*

$$\begin{aligned} F(\alpha|a|\beta) - F(\beta|a|\alpha) &= (\alpha - \beta)S^-(a), \\ \frac{1}{2}(F(\alpha|a|\beta) + F(\beta|a|\alpha)) &= F(a) + \frac{1}{2}(\alpha + \beta)S^+(a) + \alpha \cdot a \cdot \beta. \end{aligned}$$

In view of Lemma 2, it follows immediately how to concatenate new elements α and β with the given sequence a, in order to decrease the value of the functional $F(\cdot)$. The following theorem is the main result in this section.

Theorem 1. *Let there $\alpha \geq 1$ and $\beta \geq 1$ be given numbers, and let there be given the sequence a related to the $F(a)$-problem. Then there hold the following* (*respective*) *statements:*

$$F(\alpha|a|\beta) \leq \text{ (resp. } \geq) \; F(\beta|a|\alpha) \;\text{ iff }\; (\alpha - \beta)S^-(a) \leq \text{ (resp. } \geq) \; 0.$$

In theorem below we give a greedy strategy for solving our $F(a)$-problem. This strategy is based on the behaviour of the signature $S^-(\cdot)$ only.

Theorem 2. *Let there $a = (a_1, \ldots, a_n)$ be a non-decreasingly ordered sequence for the $F(a)$-problem, $u = (a_1, \ldots, a_{k-1})$, $\alpha = a_k$, $\beta = a_{k+1}$, where $1 < k < n$ and $\alpha \leq \beta$. Then there hold the following implications:*

$$\text{if } S^-(u) \geq \text{(resp. } \leq) \; 0, \;\text{ then }\; F(\alpha|u|\beta) \leq \text{(resp. } \geq) \; F(\beta|u|\alpha).$$

3 Greedy Algorithm

Let u denote a V-sequence composed from the first $k \geq 1$ elements of ordered non-decreasingly sequence a, with the actual value of functional $F(\cdot)$ equal

to $F(u)$. Let $\alpha = a_{k+1}$ and $\beta = a_{k+2}$ be two consecutive elements of a. Clearly, $\alpha \leq \beta$. Then letting $u^L = (\alpha|u|\beta)$ and $u^R = (\beta|u|\alpha)$ we obtain two possibilities to extend simultaneously the left and the right branch of the V-sequence u by the concatenation α on the left and β on the right, or conversely. On the basis of this observation we can formulate the following algorithm for the problem.

Greedy Algorithm for the Problem $1|p_j = 1 + \alpha_j t|\sum C_j$

Input: the sequence $\hat{a} = (a_0, a_1, a_2, \ldots, a_n)$, where $a_j = 1 + \alpha_j$ for $j = 0, 1, 2 \ldots, n$
Output: the sub-optimal $V-$sequence $u = (a_0, u_1, u_2, \ldots, u_n)$ of the sequence $\hat{a}$

Step 1. { `Initialization` }
Sort the sequence $\hat{a}$ in non-decreasing order, $a_{[1]} \leq a_{[2]} \leq \cdots \leq a_{[n]} \leq a_0$;
Step 2. { `Main loop` }
If n is odd **then**
begin
$u := (a_{[1]})$
for $i := 2$ **to** $n-1$ **step** 2 **do**
if $S^-(u) \leq 0$ **then** $u := (a_{[i+1]}|u|a_{[i]})$ **else** $u := (a_{[i]}|u|a_{[i+1]})$
end
else { n `is even` }
begin
$u := (a_{[1]}, a_{[2]})$
for $i := 3$ **to** $n-1$ **step** 2 **do**
if $S^-(u) \leq 0$ **then** $u := (a_{[i+1]}|u|a_{[i]})$ **else** $u := (a_{[i]}|u|a_{[i+1]})$
end
Step 3. { `Final sequence` }
$u := (a_0|u)$.

Example 1. Let $a = (2, 3, 4, 6, 8, 16, 21)$. The optimal V-shaped sequence is $a^\star = (21, 8, 6, 3, 2, 4, 16)$, with $\sum C_j(a^\star) = 23226$. The algorithm gives the V-shaped sequence $u = (21, 8, 6, 2, 3, 4, 16)$, with the value $L(u) = 23240$. Other heuristics, e.g. H_1 and H_2 proposed in [11], give worse V-sequences: $a_{H_1} = (21, 8, 4, 2, 3, 6, 16)$, $a_{H_2} = (21, 6, 3, 2, 4, 8, 16)$, with $\sum C_j(a_{H_1}) = 23418$ and $\sum C_j(a_{H_2}) = 24890$, respectively.

Thus, in general, the algorithm is not optimal. We conjecture, however, that this algorithm is optimal for *arithmetic* sequences.

Example 2. Let $a = (2, 3, 4, 5, 6, 7, 8)$. The algorithm generates the optimal V-shaped sequence $a^\star = (8, 6, 5, 2, 3, 4, 7)$ with $\sum C_j(a^\star) = 27494$, and solutions obtained by algorithms H_1 and H_2 are $a_{H_1} = (8, 6, 4, 2, 3, 5, 7)$ and $a_{H_2} = (8, 6, 4, 2, 3, 6, 7)$, with $\sum C_j(a_{H_1}) = 27519$ and $\sum C_j(a_{H_2}) = 28131$, respectively.

We think also that one can construct an optimal schedule knowing only the form of the sequence a, *without* calculations of the signature $S^-(a)$. The justifications of these two conjectures are given in the next section.

4 The Case of Natural Coefficients

We begin with a formula that can be easily derived from definition of signature $S^-(a) = M(\bar{a}) - M(a)$, where $a = (a_1, \ldots, a_n)$. Emphasize that if the sequence a is fixed, we will write S_n^- instead of $S^-(a)$:

$$S_n^- = \sum_{i=1}^{m}(a_1 \cdots a_i) - \sum_{i=1}^{n-m}(a_{n-i+1} \cdots a_n) + \sum_{i=1}^{n-m}(a_1 \cdots a_{m+i}) - \sum_{i=1}^{m}(a_i \cdots a_n),$$

where $1 \leq m \leq n$.

From this formula one can obtain the following useful representation of the signature in the case $n = 2m$ (resp. $n = 2m - 1$).

Lemma 3. *Let $a = (a_1, \ldots, a_n)$. If $n = 2m$ then*

$$S_{2m}^- = \sum_{i=1}^{m} \eta_i(m) \left(a_1 \cdots a_{m-i+1} - a_{m+i} \cdots a_{2m}\right),$$

where $\eta_1(m) \equiv 1$ and $\eta_i(m) = 1 + (a_{m-i+2} \cdots a_{m+i-1})$ for $i = 2, 3, \ldots, m$. If $n = 2m - 1$ then

$$S_{2m-1}^- = \sum_{i=1}^{m-1} \omega_i(m) \left(a_1 \cdots a_{m-i} - a_{m+i} \cdots a_{2m-1}\right),$$

where $\omega_i(m) = 1 + (a_{m-i+1} \cdots a_{m+i-1})$ for $i = 1, 2, \ldots, m - 1$.

Let us consider sequences a related to our $L(a)-$problem by the formulae:

$$\begin{aligned} a &= (\ r_m + (-1)^m, \ldots, r_2 + 1, r_1 - 1, r_1, r_2, \ldots, r_m) \ \text{ for } n = 2m, \\ a &= (\quad s_{m-1} + 2, \ldots, s_2 + 2, s_1 + 2, s_1, s_2, \ldots, s_m) \ \text{ for } n = 2m - 1, \end{aligned}$$

where

$$\begin{aligned} r_k &= 2k - \tfrac{1}{2}((-1)^k + 3) + 1, \ k = 1, 2, \ldots, m \ \text{ for } \ n = 2m, \\ s_k &= 2k - \tfrac{1}{2}((-1)^k + 3), \qquad k = 1, 2, \ldots, m \ \text{ for } \ n = 2m - 1. \end{aligned}$$

We will refer to r_k and s_k sequences and to the related sequence a as to the *even* and *odd* sequence, respectively.

Applying Lemma 3 for the sequence $a = (a_1, \ldots, a_n)$ in the case $n = 2m$, we obtain the following formula for the signature $S_{2m}^- \equiv S^-(a)$.

Lemma 4. *For $m \geq 1$ there holds the representation*

$$S_{2m}^- = \sum_{i=1}^{m} \eta_i(m) \ \left[(r_i + (-1)^i) \cdots (r_m + (-1)^m) - r_i \cdots r_m\right],$$

where $\eta_1(m) \equiv 1$ and $\eta_i(m) = 1 + (r_1 \cdots r_{i-1})[(r_1 - 1) \cdots (r_{i-1} + (-1)^{i-1})]$ for $i = 2, \ldots, m$.

Now, on the basis of this formula, we can state our first main result in this section.

Theorem 3. *Let $n = 2m$ and let $a = (a_1, \ldots, a_n)$ be an* even *sequence. Then for signatures S^-_{2m+2} and S^-_{2m} the following formula holds true*:

$$S^-_{2m+2} = r_{m+1} S^-_{2m} + (-1)^{m+1} \sum_{i=1}^{m+1} \eta_i(m)\ (r_i + (-1)^i) \cdots (r_m + (-1)^m),$$

where signature S^-_{2m} and coefficients η_i are defined as in Lemma 4. Moreover, the following identity holds true:

$$S^-_{2m+2} = R_m[(-1)^{m+1} + \Theta_m],$$

where $\Theta_m = \frac{S^-_{2m}}{R_m}(r_{m+1} + (-1)^{m+1})$ and $R_m = \sum_{i=1}^{m+1} \eta_i(m)\ r_i \cdots r_m$.

In a similar way one can consider the case of an odd sequence. Namely, applying Lemma 3 we obtain an explicit formula for signature $S^-_{2m-1} \equiv S^-(a)$.

Lemma 5. *For $m = 2m - 1$ there holds the following representation for signature S^-_{2m-1} :*

$$S^-_{2m-1} = \sum_{i=1}^{m-1} \omega_i(m)\ [(s_i + 2) \cdots (s_{m-1} + 2) - s_{i+1} \cdots s_m],$$

where $\omega_i(m) = 1 + (s_1 \cdots s_i)(s_1 + 2) \cdots (s_{i-1} + 2)$ for $i = 1, \ldots, m - 1$.

On the basis of the formula from Lemma 5 we can state our second main result in this section, concerning behaviour of signature S^-_n in the odd case, i.e.

Theorem 4. *Let $n = 2m+1$ and let $a = (a_1, \ldots, a_n)$ be an* odd *sequence. Then for signatures S^-_{2m+1} and S^-_{2m-1} the following formula holds true*:

$$S^-_{2m+1} = (s_m + 2)\ S^-_{2m-1} + (-1)^{m+1} \sum_{i=1}^{m} \omega_i(m)\ s_{i+1} \cdots s_m,$$

where $\omega_i(m) = 1 + (s_1 \cdots s_i)(s_1 + 2) \cdots (s_{i-1} + 2)$ for $i = 1, \ldots, m$. Moreover, the following identity holds true for:

$$S^-_{2m+1} = Q_m[(-1)^{m+1} + \Gamma_m],$$

where $\Gamma_m = \frac{S^-_{2m-1}}{Q_m}(s_m + 2)$ and $Q_m = \sum_{i=1}^{m} \omega_i(m)\ s_{i+1} \cdots s_m$.

Conducted computational experiments show that the term Θ_m from Theorem 3 satisfies the inequality $|\Theta_m| \leq 0.8$ for $m = 1, 2, \ldots, 1000$. Similarly, for the term Γ_m from Theorem 4 we have $|\Gamma_m| < 0.73$ for $m = 1, 2, \ldots, 1000$. Hence, it follows that signs of the signatures S^-_{2m} and S^-_{2m-1} vary according to formula $\mathrm{sign}(S^-_{2m}) = \mathrm{sign}(S^-_{2m-1}) = (-1)^m$ for $m = 1, 2, \ldots, 1000$. Moreover, the sequences a described by formulae for r_k and s_k sequences are optimal for $n = 1, 2, \ldots, 10$. These observations lead us to the following two conjectures.

Conjecture 1. The sign of signatures S^-_{2m} and S^-_{2m-1} varies according to the formula $\mathrm{sign}(S^-_{2m}) = \mathrm{sign}(S^-_{2m-1}) = (-1)^m$, where $m \geq 1$.

Conjecture 2. The algorithm formulated in Section3 generates optimal solutions for the $F(a)$-problem in the case when $a_j = j + 1$ for $j = 1, 2, \ldots, n$.

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